g. No. | | | | | | | | | | |

[22MSCH1H03]

FIRST SEMESTER UNIVERSITY EXAMINATION, JANUARY 2023

PHYSICAL CHEMISTRY-I

Time: 03 HOURS **Maximum Marks: 50**

SECTION – A

1. Answer any FOUR of the following questions. **4×5=20**

a. Define and explain Henry's law and Roult's law.

b. Prove [A, BC] = [A, B] C + B [A, C], where A, B and C are operators.

c. Derive zero order kinetics and its half-life.

d. Write and explain Onsagar conductance equation.

e. Write the Ilkovic equation for an amperometric titration.

SECTION – B

2. Answer any TWO of the following questions. **2×9=18**

a. Explain Maxwell relation with respect to

i. Combined form of I and II law of thermodynamics and

ii. Enthalpy.

b. Explain free energy and enthalpy of mixing.

c. Write and explain the Debye-Huckel-Onsager conductance equation for uni-univalent electrolyte.

SECTION – C

3. Answer the following question. **12**

a. i. Explain kinetic current in polarography. **04**

ii. Find the rate of parallel reaction. **04**

iii. State any four postulates of quantum mechanics. **04**

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